CE / CS / SE 3354 · SOFTWARE ENGINEERING · FALL 2026

Project Deliverable P1

Proposal and Requirements — Team Template

Weight 4% · Due Sunday 18 October 2026, 11:59 PM Central · Submit one PDF per team on Canvas

Field	Your entry
Project title	Sprinkler Tracking System
Team number and section	Team 02, Section 010
Team members (name · net ID)	Mujtaba Altameemi - MKA230003 Eric Dantas - EOD210001 Lucas Igl - LQI230000 Owen Shen Ash Thanickel - DAT200003
Repository URL	https://github.com/LucasIgl/CS3354.010_FALL26_TEAM02
Document version	0.1
Date	10/18/2026

0 Revision History and Contribution Record

Both tables below are graded. The contribution record carries the individual 20% of this milestone and must be consistent with your repository history.

Version	Date	Author	Summary of change
0.1	10/18	All	Initial baseline for P1

0.1 Individual contribution record

Team member	What this person produced for P1	Evidence in the repository
Mujtaba Altameemi	Section 2, 6	Document Edit History
Eric Dantas	Section 4, 6	Document Edit History
Lucas Igl	Section 1, 5, 6	Document Edit History
Owen Shen	*None*	
Ash Thanickel	Section 3, 6	Document Edit History

1 Introduction

1.1 Purpose of this document and of the system

This document acts as the proposal and baseline for Section 010, Team 02's project Sprinkler Tracking System. It contains a description of the system, its scope, context, stakeholders, requirements, and constraints. This document is used by the course instructor/grader for grading and reference, and by the team members for building, designing, and testing the system. Once signed and submitted, it becomes the baseline and additional changes must go through change control.

The Sprinkler Tracking System (STS) addresses an issue at UT Dallas where the campus wide sprinkler watering system activates on a fixed schedule regardless of the weather conditions, as observed by Team 02 members throughout the Fall 2026 semester. This results in sprinklers activating during or immediately after rain showers, causing wasted water, funds, and potentially damaging plants due to overwatering. The problem affects UT Dallas Facilities Management, since the unnecessary watering costs money and additional wear on the watering system, requiring additional upkeep. Additionally, because UT Dallas is a large campus, reducing unnecessary water use also lowers its demand on the surrounding community's water supply. The STS will provide a dashboard for the Facilities team, allowing them to easily determine when to disable the watering system's scheduled activations.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Dashboard to quickly view watering schedules and zones.	Actual rain measurement. This requires an additional hardware system with its own development and test cycle and manufacturing constraints.
Utilize weather APIs to determine when watering schedules should be disabled.	Modify the existing sprinkler system. This requires a large cost to replace a system that is not broken on its own.
Display cost and water savings from the disabled waterings.	Monitor for failing irrigation. This requires a network of additional sensors, and the goal of this project is to reduce water usage on available hardware.
Communication with the Primary Sprinkler Irrigation Controller.	Create a public-facing interface. The data collected is intended solely for UT Dallas staff and maintenance contractors.

Why the exclusions were excluded:

The exclusions are chosen since they require either a large increase in scope or cost for little gain. The goal of this project is to exist in harmony with current systems.

1.3 Definitions, acronyms and abbreviations

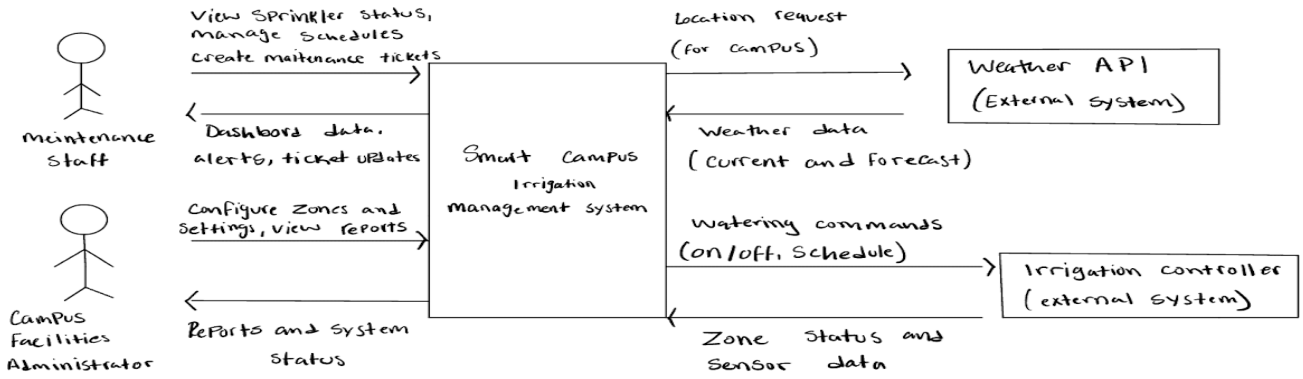
Term or abbreviation	Meaning as used in this project
STS	Sprinkler Tracking System, the name of the project being proposed in this document

1.4 References

[NA]

2 Overall Description

2.1 Product perspective and context



The Sprinkler Tracking System is designed to help campus maintenance staff keep track of the sprinklers, change watering schedules, and report any problems. It also uses weather data to check if watering is needed or not, especially when it rains. Our system will focus on the dashboard, watering schedules, and maintenance reports. The Weather API and sprinkler controller will be outside the system since we are not building those ourselves.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Maintenance Staff	Maintain the sprinklers	Check sprinkler status, change schedules, and report problems	Assumed based on their job
Campus Facilities	Manage the sprinkler system	Control watering schedules and check water usage	Assumed based on their responsibilities
University Management	Manage campus expenses	Reduce water waste and save money	Assumed based on campus needs
Students and Staff	Use campus areas	Avoid sprinklers running when they are not needed	Based on our observations on campus
IT Staff	Support the system	Keep the system secure and running properly	Assumed based on their responsibilities

2.3 User characteristics

Maintenance Staff: They know how the sprinklers work but might not be familiar with the dashboard. The system should be easy to use on phones and computers so they can check sprinkler status and report issues.

Campus Facilities Admin: They are familiar with how to manage campus operations but might not be familiar with the new system. The dashboard should be simple so they can quickly check water usage and manage watering schedules.

2.4 Constraints

1. The system will use weather APIs to determine whether watering is needed, rather than adding new sensors.
2. The system will work with the existing sprinkler system on campus without making any changes to the hardware.
3. The dashboard will only be accessible to campus maintenance staff and other authorized workers.

Source: Section 1.2 (Project Scope)

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	We are assuming we can connect our system to the sprinklers on campus.	If it doesn't work we won't be able to control the sprinklers from the dashboard.
A2	We will use a Weather API to get weather updates.	If the API doesn't work we won't be able to adjust the watering schedules based on the weather.
A3	We assume maintenance staff will be able to use the dashboard.	If they can't use it, they won't be able to check the sprinklers or change the schedules.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible, needs testing	We can use a weather API, but we haven't yet checked whether we can connect to the campus sprinklers.
Operational	Feasible, needs confirmation	Maintenance staff can use the dashboard to check the sprinklers and change the watering schedules.
Schedule	Feasible, needs planning	There are five of us in the group, so we can each work on different parts of the project.
Economic / resource	Feasible for a prototype	We are using the sprinklers already on campus, so we don't need new sensors. We still need to check the API cost.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-01	The system shall retrieve weather data for the UT Dallas campus from the weather API at least once hourly	MUST	University Management (2.2)	A tester can confirm this by checking the retrieval log
FR-02	The system shall activate the sprinklers based on the set watering schedule	MUST	Campus Facilities (2.2)	A tester can confirm this by comparing the activation of the sprinklers to the set watering schedule
FR-03	The system shall send a skip recommendation to the dashboard for the scheduled watering time if	MUST	Campus Facilities (2.2)	A tester can confirm this by simulating an acceptable forecast and approving the skip recommendation sent

ID	Requirement — one "shall" sentence	Priority	Source	Acceptance criterion
	the weather forecast provides adequate precipitation			
FR-04	The system shall allow an administrator to make modifications to the watering schedule for each watering zone	MUST	Campus Facilities (2.2)	A tester can confirm this by creating, editing, and deactivating zones and confirm the changes on the dashboard
FR-05	The system shall display the watering schedules, status, and alerts for each zone	MUST	Maintenance Staff (2.2)	A tester can confirm this by simulating alert triggers and checking the dashboard
FR-06	The system shall require a user to sign in before displaying the dashboard	MUST	IT Staff (2.2)	A tester can confirm this by requesting the dashboard and ensuring that the dashboard redirects and verifies sign in data appropriately
FR-07	The system shall allow maintenance staff to report problems with zones that will be displayed on the dashboard	SHOULD	Maintenance Staff (2.2)	A tester can confirm this by submitting a report and ensuring that it appears on the dashboard
FR-08	The system shall export watering history and skipped watering as a CSV for a user-selected range of dates	COULD	University Management (2.2)	A tester can confirm this by attempting an export with a set of known records to ensure all incidents are exported correctly

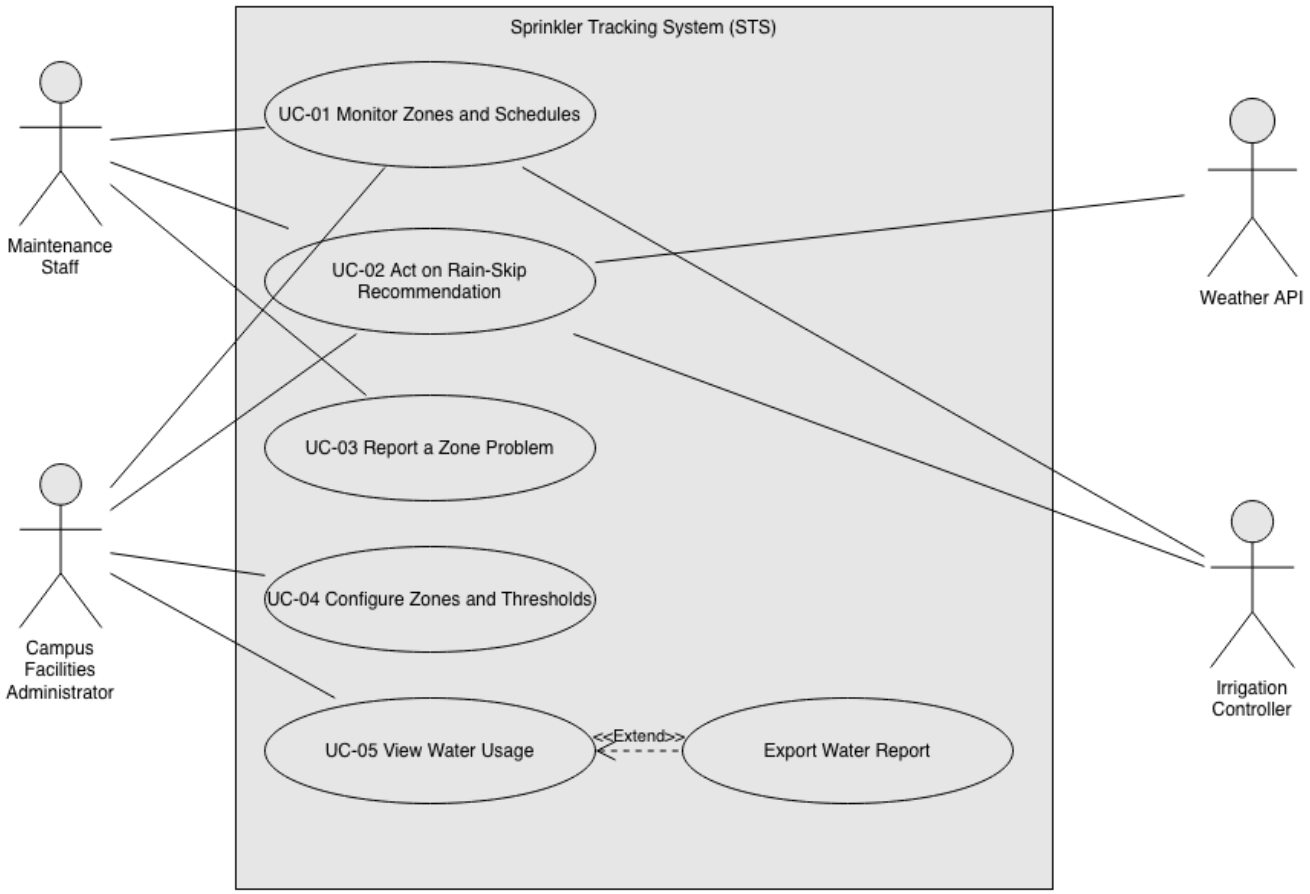
3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	Ninety percent of dashboard loads shall occur within 3 seconds, measured over a 30-minute load test with up to 20 concurrent users	MUST	Maintenance Staff, Campus Facilities (2.2)	Load test scripted against the dashboard and report the 90th percentile load time
QR-02	The dashboard shall be available for ninety-five percent of 5-minute checks between 6 AM and 10 PM over a 24-hour period	MUST	Campus Facilities (2.2)	Test script to ping the dashboard every five minutes during the period and report the successful checks
QR-03	At least 4 of 5 first-time users shall approve a rain-skip recommendation within 2 minutes without help, using only the dashboard	MUST	User Characteristics (2.3)	Timed usability session with 5 new participants
QR-04	A signed-in session shall expire after 30 minutes of inactivity and 100% of data endpoints shall reject unauthenticated requests	MUST	IT Staff (2.2)	Automated test that calls all endpoints without a session and with a session idle for 31 minutes

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall obtain all weather and rainfall data from an external weather API	Scope 1.2	New rain-measurement hardware is out of scope
CON-02	The system shall be accessible only to authenticated UT Dallas Facilities staff and authorized maintenance contractors with no public-facing interface	Scope 1.2	Zone locations and schedules are operational data only for campus staff, a public interface is out of scope

3.4 Initial use cases



Expanded use cases

Field	Content
Identifier and name	UC-01 Monitor Zones and Schedules
Primary actor	Maintenance Staff
Supporting actors	Irrigation Controller, Weather API
Preconditions — what must be true before it starts	The staff member is signed in, and at least one active zone is configured.
Trigger — what starts it	A staff member opens the dashboard.
Main success scenario — numbered steps	1. The staff member opens the dashboard 2. The system shows every active zone with its name, location, next scheduling watering, and rain-skip status, plus the time of the last weather retrieval 3. The staff member selects the zone 4. The system requests that zone's status from the Irrigation Controller 5. The Irrigation Controller returns the status 6. The system shows the zone's status (idle, watering, or offline) with the time received, next week's watering schedule, and open problem reports 7. The staff member now knows all status of active zones, use case ends
Alternative and exception flows	2a. No zones are active: The system shows "No active zones" and the use case ends.

Field	Content
	2b. The last weather retrieval was over two hours old: System shows stale weather warning and marks status as out of date. 5a. The controller does not respond within 10 seconds: the system shows last known status with timestamp with label “controller unreachable” and continues to step 6 6a. The staff member sees a fault in the zone: they may start UC-03 Report a Zone Problem for that zone
Postconditions — what is true when it succeeds	The staff member has seen the zone’s status and schedule. No data in the system has changed
Requirements realized	FR-01, FR-05, FR-06, FR-07

Field	Content
Identifier and name	UC-02 Act on Rain-Skip Recommendation
Primary actor	Campus Facilities Administrator
Supporting actors	Weather API, Irrigation Controller, Maintenance Staff
Preconditions — what must be true before it starts	The administrator is signed in. At least one active zone has a flow rate, a rain threshold, and scheduled watering. Weather retrieval has been successful within the last 2 hours.
Trigger — what starts it	The system generates a rain-skip recommendation, and the administrator sees it on the dashboard
Main success scenario — numbered steps	1. The administrator opens the recommendation on the dashboard 2. The system shows the zone, the scheduled watering time, the observed and forecast rainfall, and the zone’s threshold 3. The administrator approves the skip 4. The system sends a skip command for that watering to the irrigation controller 5. The irrigation controller acknowledges the command 6. The system records the skip with approver and time 7. The system marks the watering as skipped on the dashboard for all users
Alternative and exception flows	3a. The administrator rejects the recommendation: the system records the rejection and leaves scheduled watering; use case ends. 3b. Watering begins before the administrator acts: the system marks the recommendation expired and use case ends. 5a. The controller does not acknowledge within 60 seconds: the system retries once; if it still fails, it shows a “skip not confirmed – disable manually” alert and records the watering as not skipped
Postconditions — what is true when it succeeds	The controller has accepted the skip; the skip is recorded with approver, time; dashboard shows the watering as skipped.
Requirements realized	FR-01, FR-03, FR-04, FR-05, FR-06

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realize it	If blank — why, and what you will do
FR-01	UC-01, UC-02	
FR-02	UC-02, UC-04	
FR-03	UC-01, UC-02	
FR-04	UC-04	
FR-05	UC-01, UC-02	
FR-06	All (Precondition)	

Requirement ID	Use case(s) that realize it	If blank — why, and what you will do
FR-07	UC-03	
FR-08	UC-05	

The matrix revealed that all our functional requirements are covered through our use cases, so few modifications are required.

4 Project Plan

4.1 Team roles and working agreement

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Mujtaba Altameemi	Use cases and traceability; P2 class and sequence diagrams	Majority vote with all participating members
Owen Shen	Description lead; Weather API controller research	Majority vote with all participating members
Lucas Igl	Document owner, repo admin, document integration	Majority vote with all participating members
Eric Dantas	Project planner, risks; P2 architecture drafts	Majority vote with all participating members
Ash Thanickel	Requirement Lead; P2 State models	Majority vote with all participating members

4.2 Plan to P2

Date	What will be finished	Owner
Oct 23	Review P1 Feedback; update and log any requirement changes through change control channels	Lucas, Eric
Oct 25	Chosen weather API, cost and rate limit checked	Mujtaba
Oct 29	Contact facilities for controller details, confirming sprinkler connection is feasible	Mujtaba
Oct 31	Stakeholder confirmation, elicitation in document	Owen
Nov 1	Draft architecture and class diagrams	Eric
Nov 6	Sequence diagrams for UC-01 and UC-02; state models for zone and rain skip recommendations	Ash
Nov 8	P2 review, check requirements and submission	Lucas

4.3 Risks

#	Risk	Likelihood	Impact	Mitigation — action and owner
R1	We're unable to connect to the campus irrigation controller system (A1)	High	High	(Mujtaba) requests the controller interface documentation, if not viable (Eric) creates a simulated controller so the development can continue
R2	Weather API is too costly, rate limited or unreliable (A2)	Medium	High	(Mujtaba) compares free API's and adds caching of the last retrieval so that hourly polling stays within limits
R3	Maintenance staff or admins find the dashboard hard to use (A3) (QR-03)	Medium	Medium	(Owen) makes a layout and tests with 3 people, fixing problems found
R4	Stakeholder needs are wrong because they were assumed (2.2)	Medium	High	(Eric) emails/interviews facilities and updates sections and requirements sources as needed
R5	Uneven workload leaves sections incomplete	Medium	Medium	(Lucas) turns plan items into a GitHub issue with assignees and the team checks progress regularly. Contributions are compared to commit history

4.4 Repository and tooling

https://github.com/LucasIgl/CS3354.010_FALL26_TEAM02

All team members have access, and merge requests will be reviewed by another member to the person who created it. Issues are used to assign work and track progress.

4.5 Declaration of AI tool use

Tool	What it was used for	Who used it	How the output was checked
Claude	Final Proofreading	Lucas	Checked all suggestions against material before implementing

5 Open Questions

#	The question	Who can answer it	Needed by	What you assumed in the meantime
Q1	Does UTD already have a repair ticketing system for grounds related issues?	UTD Facilities	P2	There is no current system or it is inadequate.
Q2	Are there multiple primary irrigation controllers?	UTD Facilities	P2	There is one primary controller, but we are designing the system to communicate with many if needed.

#	The question	Who can answer it	Needed by	What you assumed in the meantime
Q3	Is there a way to directly communicate with existing irrigation controllers?	UTD Facilities	P2	There is a direct way to connect over the internet, or we can notify an on duty employee to disable the scheduled watering.

6 Baseline Declaration

We confirm that the requirements in section 3 represent our agreed baseline as of the submission date, and that every team member has read this document in full.

Team member	Signature or typed name	Date
Mujtaba Altameemi	Mujtaba Altameemi	10/18/2026
Eric Dantas	Eric Dantas	10/18/2026
Lucas Igl	Lucas Igl	10/18/2026
Owen Shen		
Ash Thanickel	Ash Thanickel	10/18/2026

How P1 Is Marked

P1 is worth 4% of the course grade. Following the syllabus, 80 points are for the shared team artifact and 20 for individual contribution evidence.

Criterion	Points	What full marks look like
Problem, scope and context (1.1, 1.2, 2.1)	10	The problem is a problem somebody actually has. The out-of-scope column is filled and the exclusions are justified. The context diagram's boundary matches what sections 2 and 3 claim.
Stakeholders and elicitation (2.2, 2.3)	10	Stakeholders include at least one who will never use the system. Every need has a stated basis, and assumptions are labelled as assumptions rather than disguised as findings.
Requirements quality (3.1, 3.2, 3.3)	25	Every requirement is one testable “shall” sentence with an identifier, a priority, a source and an acceptance criterion. Quality requirements carry a threshold, a condition and a method. Constraints are distinguished from quality requirements.
Use cases and traceability (3.4, 3.5)	20	Use cases are at actor-goal level, not steps. Two are written out in full, including alternative flows. The matrix is complete and the team says what the gaps mean and what they did about them.
Feasibility and plan (2.6, 4)	10	Each feasibility verdict is backed by evidence a reader can check. The plan has dates and owners. Risks are future events with actionable mitigations.
Open questions and baseline (5, 6)	5	Real open questions, with an owner and a date, and an honest statement of what was assumed in the meantime.
Document quality	—	Guidance boxes removed, sections in order, within the length budget, readable. Marks are deducted here, not awarded.
Individual contribution evidence (0.1)	20	Every member's contribution is distinguishable and traceable to something in the repository a reader can open.

▸ **WHERE MARKS ARE LOST**

Three things that cost teams the most marks every year:

1. Requirements that cannot be tested. “The system shall be user-friendly” earns nothing, however good the rest of the document is. Section 3 is 25 of the 80 team points and this is where they are lost.

2. Use cases written at step level. “Log In”, “Validate Card” and “Open Gate” are three steps of one use case. A diagram of fifteen tiny ovals scores lower than one with five real ones.

3. An empty section 5. It reads as a team that resolved its ambiguities by guessing.